

Gilbert Chang

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Education

Purdue University – BS in Mechanical Engineering, BS in Computer Science (Honors)
Minor in Mathematics
Fundamentals of Engineering (FE), Mechanical — Passed (NCEES), Dec 2025

Aug. 2023 – May 2027
GPA: 3.72/4.00

Research Interests

I am interested in the computational design of robotic systems: optimizing morphology, actuation, and sensing across high-dimensional design spaces, with an emphasis on biomimetic and compliant mechanisms, as well as joint optimization with robot controllers. I'm particularly interested in using hardware-accelerated differentiable high-fidelity simulations to rapidly search design spaces, but in hard-to-model regimes (e.g., contact-rich, multiphysics, aeroelastic), I'm also interested in exploring the usage of learned, physics-informed dynamics models grounded in experimentation.

Research Experience

Undergraduate Researcher, CoMMA Lab

Aug. 2024 – Present

- Developed [diff-vbd](#), an open-source differentiable implementation of the vertex block descent method for simulating hyperelastic bodies with intersection-free incremental potential contact; fit hyperelastic constitutive laws (Neo-Hookean, Mooney-Rivlin, Input Convex Neural Network) to noisy force trajectories extracted from tensile tests (Mark-10 F1505) of Formlabs Silicone 40A test samples, via inverse differentiable simulation.
- Authored [yam_rt](#), an open-source real-time C++ control library for the I2RT YAM robot arm; direct joint-space control of the arm and gripper over a SocketCAN transport, backed by a rigid-body dynamics model (derived via refinement of the analytical model with system identification experiments), raising stable control bandwidth from 100 to 500 Hz.
- Co-designing information optimal capacitive tactile sensors by coupling [diff-vbd](#) with an electrostatic capacitance imaging model, optimizing surface textures for Fisher information via implicit differentiation for design gradients.
- Developed an imitation-learning pipeline pairing a SIMD-accelerated motion planner as a real-time DAGger expert with a 25M-parameter causal-transformer policy and PointNet++ encoder, training a Franka Panda to generalize across randomized obstacle fields in PyBullet.
- Designed a soft-gripper test bench with a custom load cell and an ESP32 / six-channel differential-pressure-transducer protoboard, achieving closed-loop force control and slip detection on fluidically innervated grippers via learned force regression.

Undergraduate Researcher, Hyun Research Group

Aug. 2025 – Jun. 2026

- Engineered a lightweight Nylon rolling-element bearing architecture to minimize servo-driven wing transmission inertia; sized contact surfaces via Hertzian contact-stress analysis for infinite fatigue life at 4–6 Hz operating frequencies.
- Optimized a hybrid carbon-fiber/PLA frame via FEA to a >9 safety factor; derived Lagrangian transmission dynamics in a custom Python simulator to predict linkage forces and fatigue life.
- Accelerated design optimization and controller design by formulating data-driven linkage kinematics (polynomial regression) for faster differentiation and system identification through Volterra series fitting to experimental data.

Undergraduate Researcher, Cai Group, Ray W. Herrick Laboratories

Jan. 2025 – May 2025

- Developed a Python/MATLAB system-identification pipeline for a psychrometric test chamber, using the CVX convex-optimization toolbox to fit open-loop experimental data to RC thermal networks for control synthesis.
- Validated transient thermal models in closed-loop chamber experiments, matching empirical trajectories to within 5 °C across the full operating envelope; coordinated validation schedules and reported findings.

Undergraduate Researcher, Purdue Mechanical Engineering (Prof. Jitesh Panchal)

Jan. 2024 – May 2024

- Ran topology optimization for racing-bicycle components and validated 3D-printed Kammtail virtual foil geometries in wind-tunnel tests against CFD drag predictions.

Professional Experience

Mechanical Engineering Intern, Prototyping, 1X Technologies – Palo Alto, CA

Jan. 2026 – May 2026

- Scaled a robotic-finger endurance test platform to 55 deployed units through iterative DFM/DFA, enabling fatigue validation at production scale; fabricated cable-tension sensors by machining micro-flexures with bonded strain gages.
- Designed a custom mixed-signal PCB (instrumentation amplifiers, analog filtering, ADC) and validated its strain regime in Ansys Mechanical, capturing high-fidelity strain data for the endurance test platform's fatigue characterization.
- Built a JAX-accelerated tactile-sensor calibration rig with a custom load-cell mount and ran rigid-body

system-identification, improving the positional accuracy of a UR3e robotic manipulator to enable repeatable sensor validation.

- Authored a C++ driver for an ATI Nano17 force sensor and a CMA-ES protocol to auto-tune adaptive Kalman filter covariances from experimental and synthetic force trajectories, reducing filter-induced phase lag by 83%.

Mechanical Engineering Intern, Persona AI – Houston, TX

May 2025 – Aug. 2025

- Designed cable-tension sensors for a cable-driven robotic hand; derived closed-form strain-voltage models validated with FEA (Ansys Mechanical) to jointly optimize flexure geometry for signal-to-noise ratio and infinite fatigue life.
- Drove robotic-hand prototype hardware from GD&T drawings (ASME Y14.5) to fabrication, specifying process-aware tolerances for MJF 3D printing and off-the-shelf fasteners to yield assembly-ready parts.
- Commissioned a modular joint-angle characterization fixture (absolute encoders and signal conditioning) to isolate sensor drift under high-interference MIG magnetic fields via targeted shielding and filtering.

Undergraduate Teaching Assistant, Purdue University ECE

May 2024 – Aug. 2024

- Led instruction and problem sessions for ECE 20001: Electrical Engineering Fundamentals I; the only TA in the cohort to hold in-person office hours to support on-campus students.

Publications and Manuscripts

- B. Fan, G. Chang, J. Lai, Z. Kingston, L. Chin. "Evaluating Barometric Tactile Sensor Designs in Simulation." Late Breaking Results, IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2026.
- S. Coumar, **G. Chang**, N. Kodkani, Z. Kingston. "Foam: A Tool for Spherical Approximation of Robot Geometry." *arXiv:2503.13704*, 2025. Code: github.com/CoMMALab/foam

Presentations

- J. Lai, G. Chang, Z. Lin, A. Niu, Z. Kingston "The Development, Assembly, and Training of a Low-cost Bipedal Robot." Poster presentation, Purdue Summer Undergraduate Research Conference, Jul. 2026.
- G. Chang, Z. Kingston. "Differentiable Inverse Finite-Element Characterization of Hyperelastic Materials with Polyconvex Neural Networks." Oral presentation, Purdue Spring Undergraduate Research Conference, Apr. 2026.
- S. Zhou, G. Chang, L. Dihel, J. Hernandez, N. Hyun. "A Bio-Inspired Gliding-Enabled Flapping-Wing Aerial Robot Platform: Design, Build, and Characterization." Poster presentation, ICON Student Research Conference, Purdue University, Jan. 2026.
- S. Zhou, G. Chang, L. Dihel, J. Hernandez, N. Hyun. "Flapping-Wing Vehicles and Avian-Inspired Robotics." Oral presentation, ICON Student Research Conference, Purdue University, Jan. 2026.
- G. Chang, B. O. Philhower, D. Than. "Chiplet Integration on PCB." Oral presentation, Aug. 2024.

Awards and Honors

- Technical Merit Award — Purdue ICON Student Research Conference, Jan. 2026.
- Kenneth Ralph Scudder Mechanical Engineering Scholarship — Purdue University School of Mechanical Engineering, Jun. 2025.

Leadership and Service

- President, Five Dynamics (student rocketry organization), Purdue University — founded and led a 30+ member team; engineered two solid-rocket-motor iterations on a limited budget using OpenMotor, FEA, and DFM/DFA. (Aug. 2024 — Jan. 2025).
- Aerodynamics & Composites Subteam, Purdue University: composite layup (VARTM/RTM), CAD, and FEA for aerostructures (Aug. 2023 – May 2025).
- Team Member, Purdue System-on-Chip Extension Technologies: chiplet-on-PCB integration in KiCad (May 2024 – Aug. 2024).
- Committee Member, Taiwanese Student Association, Purdue University (Aug. 2024 – May 2025).
- Technical Volunteer, VEX Girl Powered: mentored students building and programming VEX competition robots (2023, 2024, 2025).

Skills

CAE: Siemens NX, SolidWorks, Onshape, Autodesk Inventor/Fusion 360; Ansys Mechanical/Fluent, Abaqus, FEMM; FEA, multiphysics simulation, GD&T (ASME Y14.5), DFMA; topology optimization (SIMP formulation)

Fabrication: CNC/manual mill & lathe; FDM/SLA/SLS/MJF 3D printing; VARTM/RTM composites, laser cutting

Testing & Instrumentation: load cells, strain gages, UTMs, oscilloscopes, signal conditioning, system identification

Hardware & Embedded: mixed-signal PCB design (KiCad), ESP32, Jetson Orin Nano, CAN/SocketCAN, I2C, SPI

Programming: Python (JAX, NumPy, PyTorch, SciPy), C/C++, MATLAB; Pinocchio, PyBullet; Git, Docker, Linux